

Cross-Generator Generalization in Deepfake Detection

Milestone: Preliminary Project Proposal

Description

The project aims to design a deep learning model to detect deepfakes by classifying false vs real data. The dataset to be used contains real and manipulated facial videos. Exploratory Data Analysis (EDA) will be performed to properly understand the data and identify the need for preprocessing before modeling. The project will train and compare two convolutional neural network (CNN) architectures: EfficientNet and XceptionNet. Appropriate evaluation metrics will be implemented to assess the effectiveness of the model and ensure the accuracy of the predictions.

This project focuses on computer vision applications, specifically the detection of AI-generated and manipulated facial video content. It is closely related to generative AI because deepfakes are created using generative techniques that synthesize or alter human faces in realistic ways. The selected topic is cross-generator generalization in deepfake detection: the goal is to examine how well deepfake detection models can identify manipulated content when the manipulation method used during testing is different from the methods seen during training. The project uses the FaceForensics++ manipulation methods (DeepFakes, Face2Face, FaceSwap, and NeuralTextures) as the generative sources.

For the detection task, the project uses pretrained CNN-based vision architectures, specifically EfficientNet and XceptionNet. These models are appropriate because deepfake detection often depends on identifying subtle spatial artifacts in facial regions, such as texture inconsistencies, blending errors, or unnatural visual patterns introduced during manipulation. Rather than training from scratch, the pretrained EfficientNet and XceptionNet backbones are adapted through transfer learning for a binary real-versus-fake classification task, which is more realistic for the project timeline since training large vision models from scratch would require significantly more data, computing power, and time. All project files, code, and implementation progress are maintained in the team's GitHub repository for version control, collaboration, and reproducibility.

Problem Statement

Deepfakes are digital forgeries created with AI that are highly realistic and difficult to identify at first glance. Research shows that less than 1% of people can effectively detect deepfake content (iProov, 2025, as cited in Zupan, 2025). This technology is dangerous as it has been used to commit unethical practices such as fraud, harassment, and the distribution of false information. Despite efforts to detect deepfakes, there is a gap between AI detection tools and the increase in

deepfakes, as this threat grows faster than the tools to detect it. As deepfakes become more sophisticated and realistic, more and better tools are needed to detect them.

Background

Modern deepfake generation was strongly enabled by Generative Adversarial Networks (GANs), which made realistic synthetic media much more possible, achieved by training two neural networks against each other. The actual term coined “deepfake” appeared in 2017, but the broader field builds on a longer history of computer vision, CGI, and media forensics research. More recently, diffusion based methods have further advanced the realism of synthetic media, which in effect has increased the urgency to address this challenge.

Approaches to detect deepfakes commonly rely on spatial cues, analyzing visual inconsistencies in individual frames, as well as temporal cues that track unnatural motion or blinking patterns across video sequences. Additional methods based on audio artifacts and physiological signals have also been explored. Models today, such as XceptionNet and EfficientNet, have shown strong performance on datasets like FaceForensics++. However, most detectors perform well mostly on known manipulation techniques and are not as reliable when exposed to unseen or newer generation methods, which leaves detection capabilities consistently trying to catch up.

Dataset description

- **Dataset name and source:** FaceForensics++ (FF++), introduced by Rössler et al. (2019). It is obtained through the project's official Terms-of-Use request form; community Kaggle mirrors of the compressed release are also available and provenance will be noted in the final report.
- **Size:** 5,000 videos in total: 1,000 real (pristine) source sequences collected from YouTube and 4,000 manipulated videos, with 1,000 fakes produced per manipulation method. We use the c23 (high-quality, H.264) release, which is roughly 20 GB combined.
- **Format:** H.264-encoded MP4 face videos, each centered on a single, largely frontal face at varying resolution (typically 480p or higher). For modeling, frames are sampled from each video and cropped to the face region to produce the images fed to the CNN detectors.
- **Data characteristics:** The same 1,000 identities are manipulated by four methods (DeepFakes, Face2Face, FaceSwap, and NeuralTextures), forming a clean design for studying cross-generator generalization: two methods are identity face-swaps (DeepFakes, FaceSwap) and two are facial reenactments (Face2Face, NeuralTextures), while two are learning-based (DeepFakes, NeuralTextures) and two are computer-graphics-based (FaceSwap, Face2Face). At the video level the data is class-imbalanced (about one real video for every four fakes), and FF++ ships an official identity-disjoint 720 / 140 / 140 train / validation / test split, so the same person never appears in more than one split.

Research questions

- Which FaceForensics++ manipulation techniques (DeepFakes, Face2Face, FaceSwap, and NeuralTextures) are most challenging for pretrained CNN-based detectors such as EfficientNet and XceptionNet to recognize, particularly when evaluated on unseen manipulation methods?
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- How much does detection accuracy drop when a CNN trained on three FaceForensics++ manipulation methods is tested on the fourth unseen method?
- Does the architecture choice (EfficientNet vs XceptionNet) affect cross-generator generalization, or do both models degrade similarly on unseen manipulation methods?

Methodology

The project will use FaceForensics++, a publicly available video-based deepfake dataset of labeled real and manipulated facial videos, as its primary corpus, with a DFDC subset reserved as an optional cross-dataset test. From there, EDA will be conducted to assess class balance, data quality, and preprocessing needs such as frame or feature extraction. During the modeling phase we will fine-tune the pretrained EfficientNet and XceptionNet architectures, which have demonstrated strong effectiveness in deepfake detection research. We will then analyze the results and identify the spatial artifacts introduced during the synthesis process. Performance and validation will be assessed using precision, recall, F1-score, and ROC-AUC, with attention to how well each model generalizes to unseen manipulations.

Expected Outcomes

The project will deliver one or more trained CNN deepfake detectors such as EfficientNet or XceptionNet, along with a reproducible preprocessing and evaluation pipeline built on FaceForensics++. The detectors will be evaluated on a held-out test split using standard metrics such as precision, recall, F1-score, and ROC-AUC, with results reported per class to account for the real-versus-fake imbalance in the dataset. Beyond in-distribution accuracy, the main contribution of the project will be a characterization of cross-generator generalization, which refers to how well each detector performs given manipulation methods it was not trained on. We will produce a transfer matrix across the four manipulation methods included in FaceForensics++ (DeepFakes, Face2Face, FaceSwap, and NeuralTextures), and analyze which methods transfer best and worst with hypotheses based on the spatial artifacts each method introduces. If time allows, we will also conduct a cross-dataset evaluation on a DFDC subset that was never used during training. The expected outcome is a measurable seen-versus-unseen accuracy gap, with attention to providing an honest characterization of the generalization limits of standard CNN detectors rather than a single top-line accuracy number.

Data Preprocessing Approach

Before the team starts with model development, an initial data exploration of the raw data will be conducted to understand its actual structure and determine which methods would be appropriate for the project's goals and objectives. This will involve examining the FaceForensics++ dataset organization and data structure and auditing how source identities are distributed and reused across the four manipulation methods (DeepFakes, Face2Face, FaceSwap, and NeuralTextures). This exploration will also include checking class balance between real and manipulated videos, since each real video corresponds to multiple fake variants, which produces an inherent imbalance that we would need to address before modeling. This initial characterization will directly shape the preprocessing approach.

Building on the exploration, the preprocessing pipeline will be structured into four stages: frame extraction, face detection and cropping, normalization and dataset splitting. Frame extraction was performed on the c23 compressed videos, with frames sampled at fixed rate intervals rather than decoding every frame, and the total number frames extracted per class capped early to maintain balance from the outset. Face detection and cropping will then be applied using a pretrained face detector to localize facial regions in each sampled frame, retaining a small margin around each bounding box to preserve context beyond the immediate facial landmarks, since certain artifacts are not always confined to the face interior. Cropped faces will be resized to a fixed resolution matching the input requirements of the EfficientNet and XceptionNet backbones and cached to disk as a reusable feature store, avoiding redundant computation across training runs. Normalization will follow standard ImageNet preprocessing conventions, consistent with the use of pretrained backbones for transfer learning, and data augmentation will be limited to horizontal flipping and mild brightness or contrast adjustments; more aggressive augmentation such as blurring or compression simulation will be avoided in order to preserve the spatial artifacts the model is intended to detect. Finally, dataset splitting will be constructed at the identity level to prevent the same individual from appearing in both training and test sets, with each manipulation method held out independently during training to support the project's cross-generator transfer matrix analysis, alongside a separate in-distribution held-out test split used to report standard classification metrics.

Model implementation:

Both detectors will be implemented in PyTorch using pretrained backbones adapted for binary (real versus fake) classification through transfer learning rather than training from scratch. For each architecture, the ImageNet-pretrained convolutional backbone is kept as a feature extractor and its original classification head is replaced with a new fully connected layer that outputs a single real-versus-fake prediction. XceptionNet takes 299x299 face crops and EfficientNet (B0/B4) takes 224 to 256 pixel crops, each normalized with the ImageNet statistics its pretrained weights expect. Training uses binary cross-entropy loss with the Adam optimizer and a low fine-tuning learning rate, together with class-weighted sampling to counter the real-versus-fake imbalance; mixed-precision training and early stopping on the validation split keep runs efficient across the team's different hardware. To make results reproducible and portable, hyperparameters and file paths are stored in version-controlled configuration files with fixed random seeds, model checkpoints are saved so long runs can resume after an interruption, and all code is maintained in the team's shared GitHub repository. The same implementation is used for both backbones so that any difference in cross-generator behavior can be attributed to the architecture itself rather than to differences in the training setup.

Experimental design:

The central experiment is a rotating leave-one-manipulation-out evaluation designed to measure directly how much detection accuracy drops on an unseen generator. For each of the four FaceForensics++ manipulation methods, a detector is trained on the real videos plus the other three methods and then tested on the fourth, held-out method, and this is repeated so that each method serves once as the unseen target. Running this protocol for both EfficientNet and XceptionNet yields eight cross-generator training runs, and each model is additionally evaluated on an in-distribution held-out split of the methods it was trained on to provide an upper-bound reference. Seen (in-distribution) and unseen (held-out) results are always reported separately rather than averaged together, and are organized into a transfer matrix whose diagonal records in-distribution performance and whose off-diagonal cells record how each training combination transfers to every other method. Comparing the two backbones on identical folds isolates

whether cross-generator robustness depends on the architecture or is a property of the conventional CNN approach itself. If time permits, a stretch experiment evaluates an FF++-trained detector on a DFDC subset that shares none of FF++'s generators or capture conditions, testing whether the generalization gap widens further under a complete change of dataset.

Evaluation metrics:

- F1-score: the harmonic mean of precision and recall, giving a single balanced measure that accounts for both false positives and false negatives.
- Precision: the ratio of correctly predicted fake samples to all predicted fake samples.
- Recall: the ratio of correctly predicted fake samples to all actual fake samples.
- ROC-AUC (Receiver Operating Characteristic - Area Under the Curve): evaluates a model's ability to distinguish between real media and fakes across entirely unseen generation methods.

Analysis techniques:

- Transfer Matrix to present the AUC for each training-testing combination.
- Plot ROC curves for EfficientNet and XceptionNet for each of the four held-out methods, displaying the in-distribution and cross-generator curves on the same axes, to visualize the trade-off between true positive rate and false positive rate and show where in-distribution and cross-generator curves diverge visually.

Timeline

- Week 1 - Exploratory data analysis and preprocessing: characterize FaceForensics++ (class balance, identity coverage across the four manipulation methods) and build the frame-extraction, face-detection, and cropping pipeline with a cached feature store.
- Week 2 - Model implementation and initial training: implement the pretrained EfficientNet and XceptionNet detectors and run initial in-distribution training with checkpointing.
- Week 3 - Cross-generator evaluation: run the leave-one-manipulation-out protocol for both backbones and assemble the seen-versus-unseen transfer matrix.
- Week 4 - Results analysis, visualization, and final report: analyze best- and worst-transfer pairs, produce ROC curves and result figures, and write the final report.

Team Roles

Every member serves as a principal researcher and independently trains and evaluates their own set of model combinations, so that the full cross-generator transfer matrix is produced in parallel and each person gains hands-on experience with the complete detection pipeline. Each member additionally owns a supporting technical area, ensuring that every stage of the project (data preparation, preprocessing, training infrastructure, evaluation, and reporting) has a clear owner. The Final Report is written collaboratively, with each member contributing the sections corresponding to their work and reviewing the work of the others.

Member	Primary Area	Specific Tasks
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Dominic Rivas	Principal Researcher	Serves as a principal researcher: independently trains and evaluates an assigned run of the model combinations, and contributes to preprocessing, evaluation, and the collaborative final report.
Jonathan Jude Regalado	Principal Researcher	Serves as a principal researcher: independently trains and evaluates an assigned run of the model combinations, and contributes to preprocessing, evaluation, and the collaborative final report.
Lyxelis Rodriguez Navarro	Principal Researcher	Serves as a principal researcher: independently trains and evaluates an assigned run of the model combinations, and contributes to preprocessing, evaluation, and the collaborative final report.
Obinna Okonkwo	Principal Researcher	Serves as a principal researcher: independently trains and evaluates an assigned run of the model combinations, and contributes to

		preprocessing, evaluation, and the collaborative final report.
Sagar Ayare	Principal Researcher	Serves as a principal researcher: independently trains and evaluates an assigned run of the model combinations, and contributes to preprocessing, evaluation, and the collaborative final report.

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